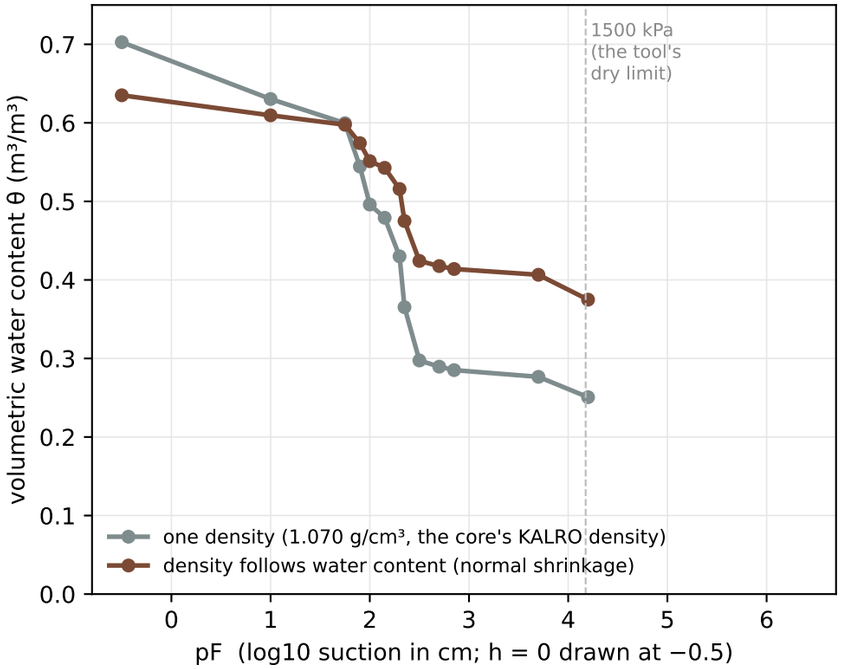
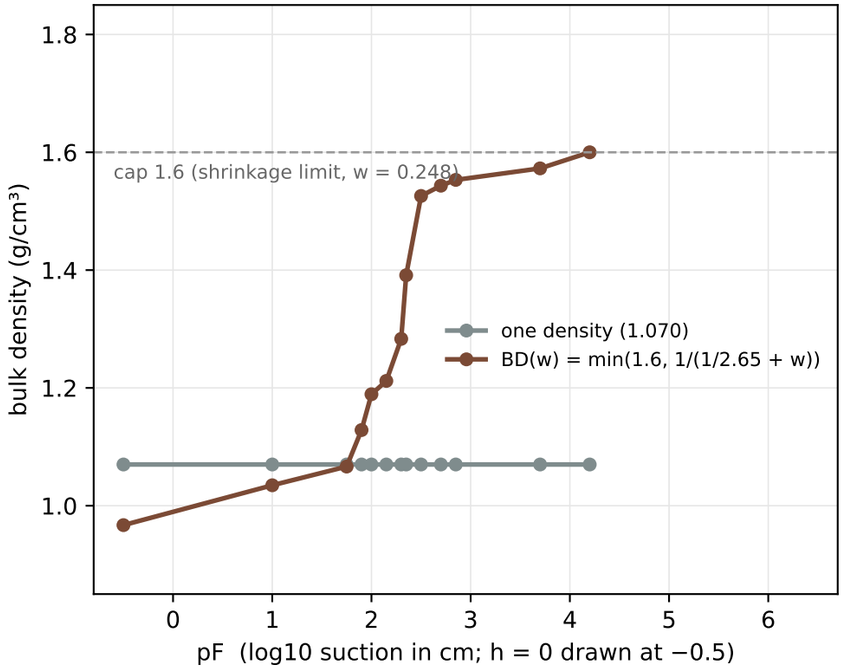
Two vertisols, two regions: converting retention curves with a shrinking bulk density

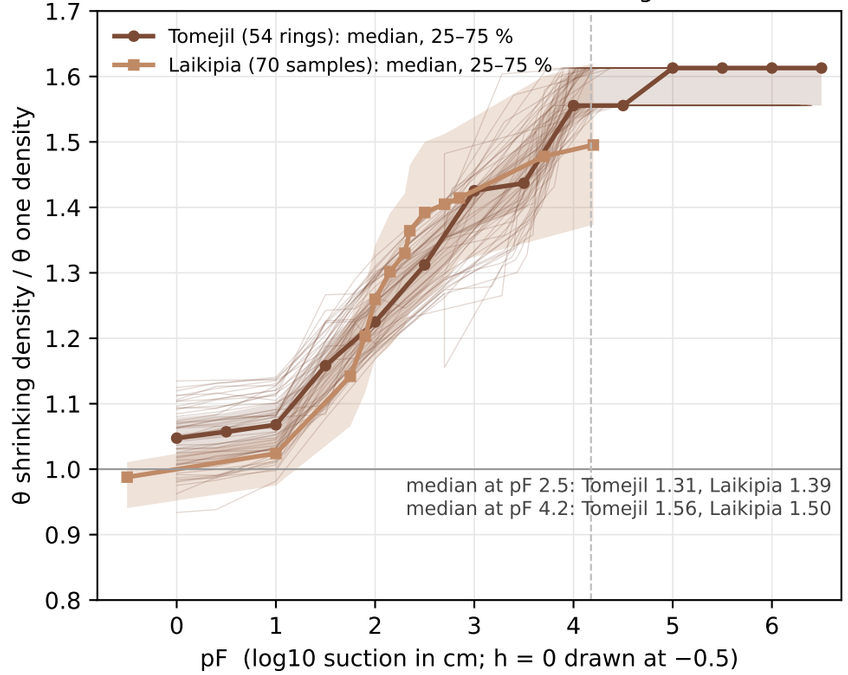
a Laikipia (Kenya), sample WL_3LT07414: retention curve



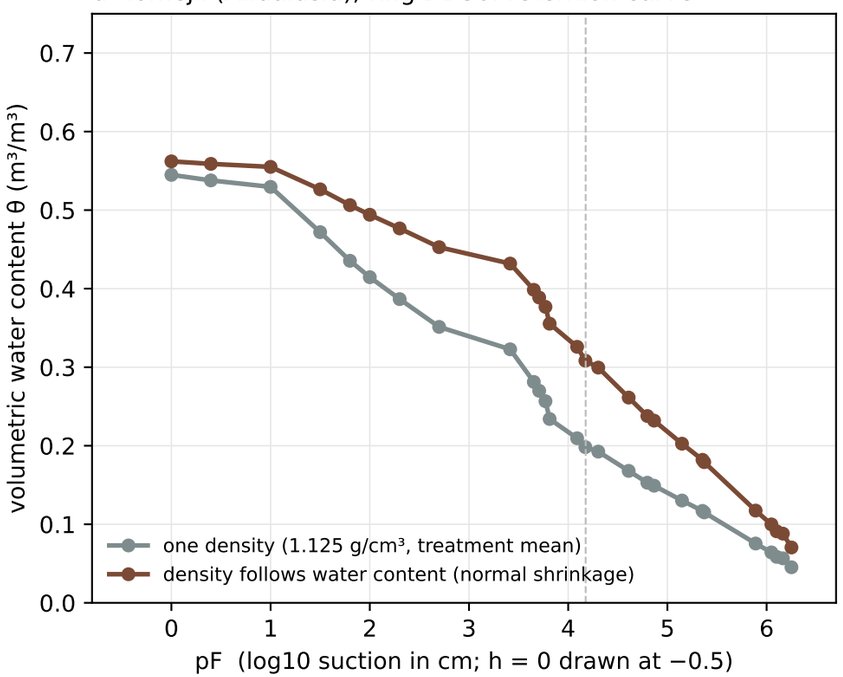
b Laikipia: density used for the conversion



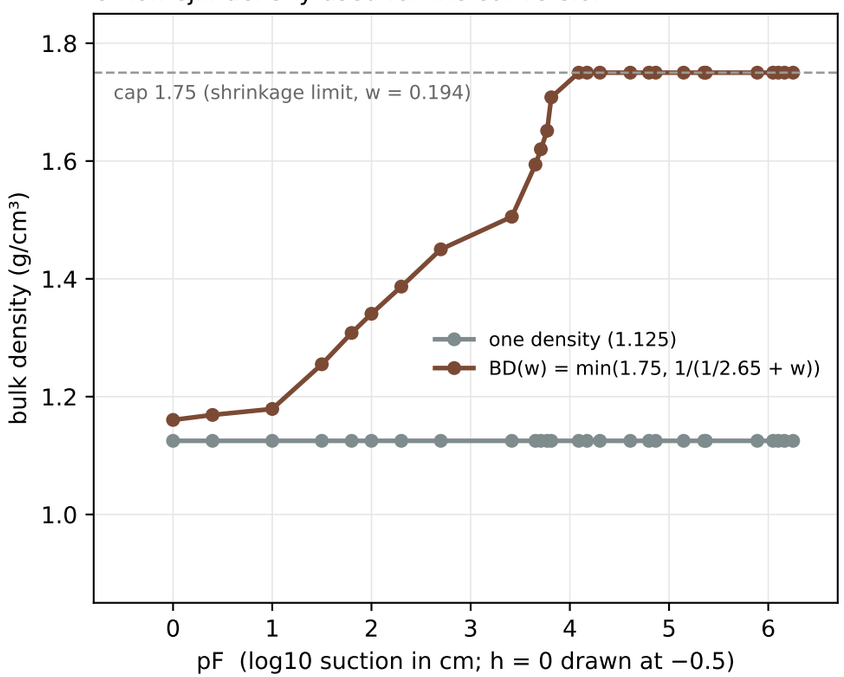
c both soils: how much the conversion changes θ



d Tomejil (Andalusia), ring DD30: retention curve



e Tomejil: density used for the conversion



f the conversion

At every suction:
  w = θ / BD (one density) gravimetric water
  BD(w) = min(cap, 1 / (1/2.65 + w)) normal shrinkage:
    the soil loses volume equal to the water it loses,
    until the shrinkage limit (the cap)
  θ = w · BD(w)

Cap and one density:
  Laikipia — cap 1.6, as in the published series (HydroShare);
    the core's KALRO density; curves published gravimetric
  Tomejil — cap 1.75, median oven-dry clod density of KSSL
    Vertisols; the treatment mean, 1.085 (conventional tillage)
    or 1.125 (direct drill; Ordóñez Fernández et al. 2007)

Where the soil is wetter than the core was dense (swollen, BD(w) below the one density), the conversion lowers θ; as it dries and shrinks, it raises θ, by about half at the dry end. Beyond pF ~3.8 most rings and cores reach their cap.

The agreement is partly built in (same formula). Independent support: with the converted curves the tool reads both soils as clay. A measured shrinkage curve or COLE would test the model directly.